

Lecture Notes on

Quantum Mechanics

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These notes cover quantum mechanics at the undergraduate level. The sources used are:

- Griffiths: *Introduction to Quantum Mechanics*

Potential sources to consult in the future:

- Sakurai: *Modern Quantum Mechanics*

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1 The Schrödinger Equation

1.1 Wave Functions

Classically, we work with masses and attempt to solve for their equations of motions $x(t)$ using corresponding Newton's laws or energy equations. The quantum mechanics approach is different; the idea is to instead solve for a particle's wave function $\Psi(x, t)$ and from there we can do anything. The goal then is to solve the Schrödinger equation

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi$$

After solving the Schrödinger equation for some initial wave function $\Psi(x, 0)$ the equation actually gives us all future values of the wave function $\Psi(x, t)$.

The statistical interpretation of the wave function goes as follows:

- Born's statistical interpretation of the wave function states that

$$\int_b^a |\Psi(x, t)|^2 dx$$

is the probability of finding the particle at $a \leq x \leq b$ at time t . Note that while Ψ is a complex valued function, $|\Psi|^2 = \Psi\Psi^*$ where Ψ^* is the complex conjugate of Ψ is always real and nonnegative.

- An obvious consequence is this is that even if you know everything about a particle (ie; you know its wavefunction), you still cannot predict with certainty the outcome of an experiment to measure its exact position.
- Before measurement, a particle's position is not well defined. Only after measurement can we say the particle has a "position," and by doing so we collapse the particle's wavefunction to a dirac delta centered around the measured position. This is known as the **Copenhagen Interpretation**
- The integral over all possible positions must be equal to one, since the particle must be located somewhere.

$$\int_{-\infty}^{\infty} |\Psi(x, t)|^2 dx = 1$$

If we have a wavefunction, we need to make sure its normalized for it to be physical. We can normalize it to obey the integral by multiplying it by a scale factor. A glance at the Schrödinger equation one can see that if $\Psi(x, t)$ is a solution, then $A\Psi(x, t)$ is also a solution for nonzero constant A .

- One can also show that the normalization integral does not change with time, so there is no need to constantly renormalized solutions as the particle evolves with time.
- If we interpret $|\Psi(x, t)|^2$ to be the probability, then we can say the expectation value of position is

$$\langle x \rangle = \int_{-\infty}^{\infty} x |\Psi(x, t)|^2 dx$$

Note that the interpretation of this quantity is the expected value of the position from measuring particles with wave function Ψ . Simply remeasuring the same particle over again will not work, since the first measurement will collapse the particle into some other wave function with exact position

- By differentiating the expected position with respect to time, one can obtain the expectation value of the velocity. It's customary for us to work using momentum so we have

$$\langle p \rangle = -i\hbar \int \left(\Psi^* \frac{\partial \Psi}{\partial x} \right) dx$$

- A more illustrative way to write the expectation values for position and momentum are:

$$\langle x \rangle = \int \Psi^* \cdot x \cdot \Psi dx$$

$$\langle p \rangle = -i\hbar \int \Psi^* \cdot \left(\frac{\partial}{\partial x} \right) \cdot \Psi dx$$

We say the the x operator "represents" position and the $-i\hbar \frac{\partial}{\partial x}$ operator "represents" momentum.

- All classical quantities can be represented as a function of position and angular momentum. Then the representation of a classical quantity $Q(x, p)$ can be written as

$$\langle Q(x, p) \rangle = \int \Psi^* Q \left(x, -i\hbar \frac{\partial}{\partial x} \right) \Psi dx.$$

- **Ehrenfest's Theorem** states that expectation values obey their classical laws. For example, the time derivative of momentum is

$$\frac{d\langle p \rangle}{dt} = \left\langle -\frac{\partial V}{\partial x} \right\rangle$$

1.2 Time Independent Schrödinger Equation

Let us assume that the potential $V(x)$ is independent of time. Then using separation of variables, we can substitute $\Psi(x, t) = \psi(x)\phi(t)$ into the Schrödinger equation giving us $i\hbar \frac{1}{\phi} \frac{d\phi}{dt} = -\frac{\hbar^2}{2m} \frac{1}{\psi} \frac{d^2\psi}{dx^2} + V$. Then for the equality to hold, both sides must be equal to some constant value E leaving us with two ordinary differential equations:

$$\frac{d\phi}{dt} = -\frac{iE}{\hbar} \phi$$

$$-\frac{\hbar^2}{2m} \frac{1}{\psi} \frac{d^2\psi}{dx^2} + V = E$$

The second equation is the **Time Independent Schrödinger Equation**. The first equation defines the characteristic time dependence and can be solved trivially $\phi(t) = e^{-iEt/\hbar}$. Separable solutions obey several important properties that we're interested in:

- These solutions are **stationary states**. What this means is while the wave function is obviously time dependent, $\Psi(x, t) = \psi(x)e^{-iEt/\hbar}$, the probability density is not:

$$|\Psi(x, t)|^2 = \Psi\Psi^* = \psi^*e^{+iEt/\hbar}\psi e^{-iEt/\hbar} = |\psi(x)|^2$$

We can further show that the expectation value of any classical quantity is also not time dependent in a stationary state:

$$\langle Q(x, p) \rangle = \int \psi^* Q \left(x, -i\hbar \frac{\partial}{\partial x} \right) \psi dx.$$

- The constant E represents the Hamiltonian of the system. Classically, the Hamiltonian is $H(x, p) = \frac{p^2}{2m} + V(x)$ so the corresponding Hamiltonian operator is $\hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x)$. So the time independent Schrödinger equation can be rewritten as

$$\hat{H}\psi = E\psi$$

Then the expectation values of the Hamiltonian is

$$\langle H \rangle = \int \psi^* \hat{H} \psi dx = \int \psi^* E \psi dx = E$$

$$\langle H^2 \rangle = \int \psi^* \hat{H}^2 \psi dx = \int \psi^* E (\hat{H} \psi) dx = \int \psi^* E^2 \psi dx = E^2$$

This means the standard deviation is 0. So any ensemble of stationary states with the same wavefunction have their Hamiltonian measured to be the exact same E .

- Every solution to the Schrödinger equation can be written in the form of a sum of stationary solutions.

$$\Psi(x, t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-E_n t/\hbar}$$

So we can basically decompose any wave function into the sum of wavefunctions associated with specific energy levels. The quantity $|c_n|^2$ represents the probability that a measurement of the energy will return value E_n , so we also must demand

$$\sum_{n=1}^{\infty} |c_n|^2 = 1$$

$$\langle H \rangle = \sum_{n=1}^{\infty} |c_n|^2 E_n$$

1.3 The Quantum Harmonic Oscillator

2 Dirac Notation

3 The Hydrogen Atom

4 Symmetries and Conservation Laws